

ZOHAIR MUQEEM

(224)-386-4197 | Chicago, IL | zohairmuqem@gmail.com

EDUCATION

University of Illinois at Urbana-Champaign

B.S. in Statistics, B.A. in Economics

Minor in Mathematics

May 2026

GPA: 3.70/4.00

Awards: Dean's List Spring 2023, Data Science Certificate, High Distinction in Statistics

Activities: Illini eSports Orange Player, Muslim Student Association, The Smashing Illini, Bee Keeping Club

TECHNICAL SKILLS

Programming & Data Tools: Python, R, SQL (Oracle SQL), Quarto, R Shiny (shinyapps.io), Streamlit, LaTeX, Excel

Statistical & Machine Learning Methods: K-Means & Hierarchical Clustering (HAC), DBSCAN, t-SNE, Maximum Likelihood Estimation, Statistical Inference & Hypothesis Testing

Data Visualization: ggplot2, gganimate, Plotly, Matplotlib, Power BI

EXPERIENCE

Interactive Brokers LLC

Chicago, IL

Model Governance Intern

June 2025 - August 2025

- Streamlined Python-based anti-money-laundering compliance scripts, enhancing execution efficiency and improving accuracy of risk-flag assignment across 4M+ brokerage accounts
- Applied statistical inference and graphical diagnostics to optimize model performance, improving the precision of flagging criteria for anti-money laundering compliance across Oracle SQL databases
- Collaborated with risk, compliance, and engineering teams to review, validate, and document AML model behavior in alignment with internal and global model governance standards, including FATF, Basel, and regional regulatory frameworks
- Competed in a \$1M simulated paper trading competition on the Interactive Brokers platform, executing options (buy/sell), Forex, and commodities futures strategies to generate over \$300K in simulated returns

University of Illinois at Urbana-Champaign

Champaign, IL

Front Desk Clerk

August 2025 – May 2026

- Tracked and logged daily package data into internal systems, maintaining accurate time-stamped records and identifying volume patterns to support package processing

Course Assistant (Statistics and Probability I)

May 2024 - August 2024

- Supported exam proctoring and weekly homework grading for 100+ students; led Zoom office hours to explain core concepts in statistical theory including Maximum Likelihood Estimation, Moment Generating Functions, and parameter hypothesis testing

Resident Advisor

August 2023 - May 2025

- Facilitated 24/7 crisis response and community building for 50 residents through housing initiatives and one-on-one conversations

Northwestern Medicine

Chicago, IL

Oncological Researcher (Hospital Internship)

August 2020 - July 2021

- Synthesized 15 medical journals into population subsets within the Chicagoland area, highlighting 10 deficient areas of care through analyzing combinations of quantitative/categorical barriers such as race, income, and gender
- Developed a strong understanding of end-of-life (EOL) patient care and medical training through simulated clinical scenarios

PROJECTS

FRED Macro-Risk Dashboard

- Built an automated data pipelining tool in R to retrieve FRED API macro-economic indicators, alongside providing an extensive data wrangling, validation, and visualization Quarto Markdown report on existing market conditions
- Deployed a live interactive R Shiny application featuring interactive ggplot2, gganimate, and Plotly charts to communicate stock volatility predictions and financial market insights using various econometric models

Global Mental Health Unsupervised Clustering Analysis

- Conducted an unsupervised machine learning analysis of global mental health disorder rates across 195 countries, implementing clustering techniques including K-Means, HAC, and various diagnostic tools such as t-SNE plots, DBSCAN, and Silhouette Heat Mapping to uncover similar mental health profiles among countries
- Designed a live Streamlit web application that streamlines clustering analysis, enabling a user-friendly interface

Oracle SQL-Python Accessor Tool

- Engineered accessor tool to extract Oracle SQL queries and establish cross-compatibility with Python, facilitating automated data extraction, cleaning, and downstream analysis